PRECALCULUS · UNIT 1: EQUATIONS AND INEQUALITIES

Concert Venue Planner

Use equations and inequalities to set ticket prices, tolerances, and capacity for a concert.

Honors

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your team is running a summer concert at Riverside Amphitheater. Ticket price affects how many people come, the sound system must stay within safe levels, and the venue has fixed costs no matter how many tickets are sold.

You will solve equations and inequalities to set the price, plan the budget, and keep the crowd safe.



Demand

At $40: 3,000 fans

−50 fans per +$1

Costs

$45,000 fixed

$6 per fan

Learning Goals

- I can model a situation with polynomial and rational functions and solve related inequalities.
- I can solve radical and absolute value equations and check for extraneous roots.
- I can justify an optimal plan with algebra.

Project Timeline

When	What you will do
Day 1	Optimal Pricing
Day 2	Average Cost & Rational Inequalities
Day 3	Sound Engineering
Final	Venue Operations Plan + reflection

Part 1: Optimal Pricing

Day 1 · n = 5,000 − 50p, cost 45,000 + 6n, capacity 4,000.

Venue Fact Cards

Capacity	Sound Limit	Stage Temp
At most 4,000 fans	Target 100 dB	Gear safe within
	within ±6 dB	5°F of 72°F

Demand	Costs	Sound Fade
n = 3,000 − 50(p − 40)	C = 45,000 + 6n	Level at distance d meters
p = ticket price		L = 112 − 4√d

1. Find the profit function and the price that maximizes profit by completing the square.

2. A sponsor adds $2 per fan. Find the new profit function and optimal price.

3. Solve P(p) > 50,000 for the original model.

4. Explain why the capacity constraint (p ≥ 20) does not change the optimal price.

Part 2: Average Cost & Rational Inequalities

Day 2 · Average cost per fan A(n) = $\frac{45{,}000 + 6n}{n}$.

5. Solve A(n) < p for the optimal price p = 53 and interpret.

6. Solve $\frac{n - 2{,}000}{n - 4{,}000} \le 0$ and give a venue meaning (e.g., comparing attendance to thresholds).

7. For which n is A(n) within $1 of its long-run value $6? Solve $|A(n) - 6| \le 1$.

8. Sketch A(n) and describe its asymptotes.

Part 3: Sound Engineering

Day 3 · Model: $L = 112 - 4\sqrt{d}$.

9. Solve $112 - 4\sqrt{(d + 5)} = 96$ and check for extraneous solutions.

10. Solve $\sqrt{(d + 7)} = d - 5$ and check. Interpret one solution.

11. Design a seating chart: VIP (≥ 100 dB), general (94–100 dB), family (≤ 94 dB). Give each zone's distance range.

Final Product: Venue Operations Plan

Your final product must include:

- Profit function in vertex form and sponsor analysis
- Profit threshold interval
- Rational inequalities with asymptotes
- Radical equations with checks
- Seating zones and the operations plan

Reflection

R1. How does vertex form reveal the optimal price immediately?

R2. Why can squaring both sides create an extraneous solution?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Equations and inequalities model price, cost, safety, and sound correctly and support the plan. Goes beyond the minimum with a creative, realistic choice.	Equations and inequalities model price, cost, safety, and sound correctly and support the plan.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Plan includes models, solution sets, a recommended price, and seating zones. Polished, organized, and visually engaging.	Plan includes models, solution sets, a recommended price, and seating zones. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.